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LAB 6: B2 SPIRIT MASS MOMENT OF INERTIA

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EXPERIMENT OBJECTIVES

- PERFORM FREE VIBRATION TESTING OF A SIMPLE GEOMETRIC SHAPE (1X3 WOOD BLOCK) TO DETERMINE ITS MASS MOMENT OF INERTIA AROUND TWO DIFFERENT AXES USING:
  - THEORETICAL METHOD (HAND CALCULATIONS)
  - TORSIONAL VIBRATION METHOD
  - PENDULUM SWING METHOD
- COMPARE THE RESULTS OBTAINED BETWEEN ALL METHODS
- PERFORM FREE VIBRATION TESTING OF A COMPLEX GEOMETRIC SHAPE (B-2 SPIRIT AIRCRAFT MODEL) TO DETERMINE ITS MASS MOMENT OF INERTIA EXPERIMENTALLY AROUND TWO DIFFERENT AXES:
  - ROLL AXIS (TORSIONAL VIBRATION)
  - YAW AXIS (TORSIONAL VIBRATION AND PENDULUM SWING)
- COMPARE THE EXPERIMENTAL METHODS FOR THE B-2 SPIRIT AIRCRAFT TO VALIDATE THE TECHNIQUE.

MASS MOMENT OF INERTIA OF WOOD BOARD (TORSION TEST SETUP)

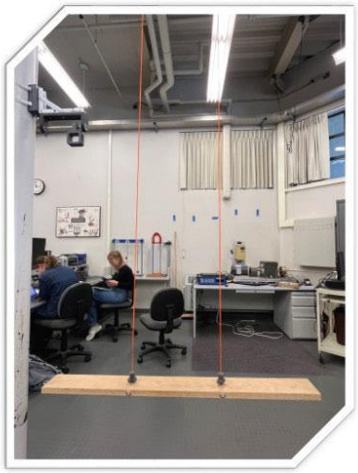
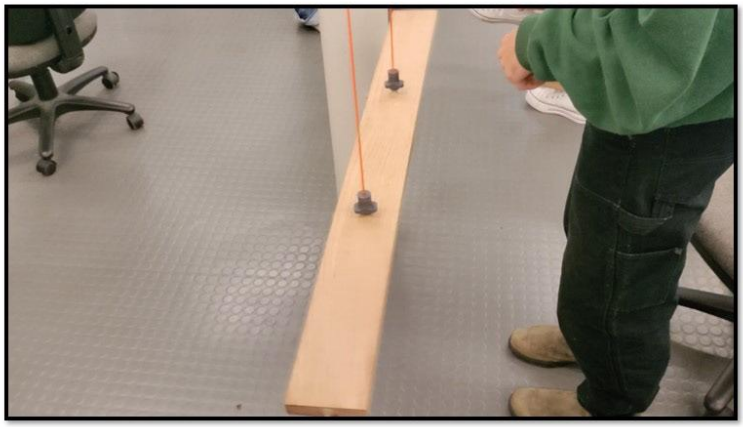


FIGURE: WOOD BOARD HUNG FROM Y AXIS



FIGURE: WOOD BOARD HUNG FROM Z AXIS

MASS MOMENT OF INERTIA OF WOOD BOARD (TORSION TEST PROCEDURE Y-AXIS)



MASS MOMENT OF INERTIA OF WOOD BOARD (TORSION TEST RESULTS Y-AXIS)

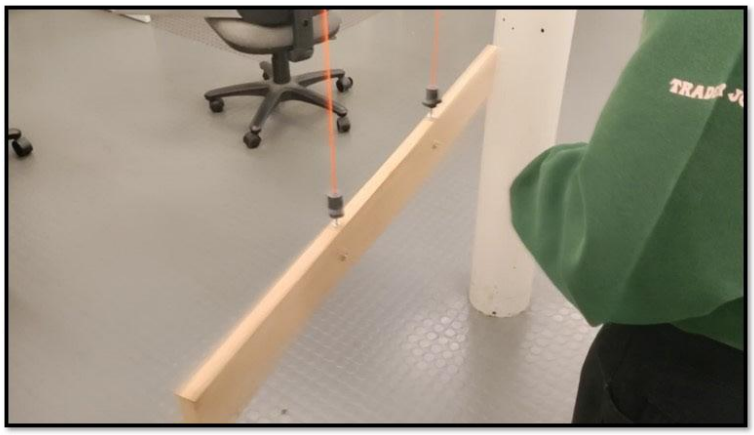
Axis or Rotation	Trial #	Duration for 5 Oscillations (s)	Average Time for 5 Oscillations (s)
Y	1	25.04	24.76
	2	24.53	
	3	24.71	

Axis or Rotation	Cable Distance (m)	Height of Cables (m)	Average Time Period (s)	Experimental MOI (kg*m^2)
Y	0.3048	2	4.95	0.0994

Axis or Rotation	Theoretical MOI (kg*m^2)	Experimental MOI (kg*m^2)	% Error
Y	0.0975	0.0994	1.98

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MASS MOMENT OF INERTIA OF WOOD BOARD (TORSION TEST PROCEDURE Z-AXIS)



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MASS MOMENT OF INERTIA OF WOOD BOARD (TORSION TEST RESULTS Z-AXIS)

Axis or Rotation	Trial #	Duration for 5 Oscillations (s)	Average Time for 5 Oscillations (s)
Z	1	24.66	24.803
	2	24.84	
	3	24.91	

Axis or Rotation	Cable Distance (m)	Height of Cables (m)	Average Time Period (s)	Experimental MOI (kg*m^2)
Z	0.3048	2	4.96	0.0998

Axis or Rotation	Theoretical MOI (kg*m^2)	Experimental MOI (kg*m^2)	% Error
Z	0.0966	0.0998	3.27

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MASS MOMENT OF INERTIA OF WOOD BOARD (PENDULUM SWING TEST SETUP)

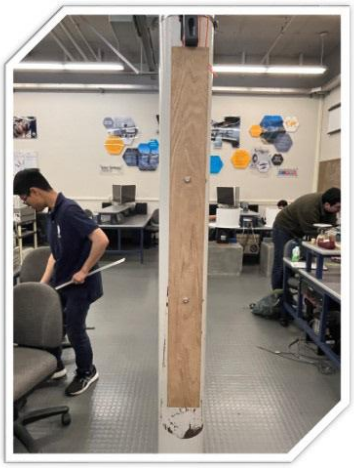


FIGURE: WOOD BOARD HUNG FROM Y AXIS

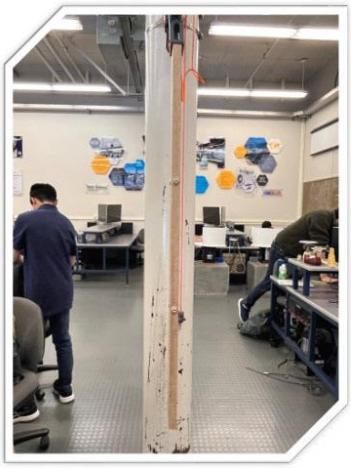


FIGURE: WOOD BOARD HUNG FROM Z AXIS

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MASS MOMENT OF INERTIA OF WOOD BOARD (PENDULUM SWING TEST PROCEDURE Y-AXIS)



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MASS MOMENT OF INERTIA OF WOOD BOARD (PENDULUM SWING TEST RESULTS Y-AXIS)

Trial #	Duration for 5 Oscillations (s)	Average Time (s)
1	8.32	8.39
2	8.42	
3	8.42	

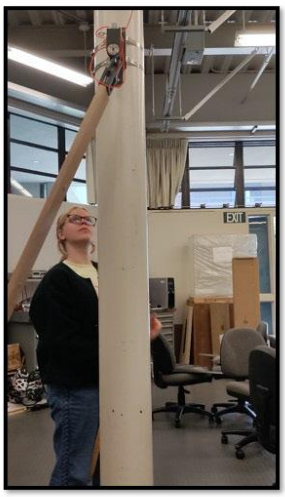
Radius to Center	Average Time Period Tavg(s)	MOI (kg*m^2)
0.535	1.68	0.125

MOI (Pendulum) (kg*m^2)	MOI (Torsional Vibration) (kg*m^2)	% Error
0.125	0.0994	25.8

Theoretical MOI	MOI (Pendulum)	MOI (Torsional Vibration)
0.0975	0.125	0.0994

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MASS MOMENT OF INERTIA OF WOOD BOARD (PENDULUM SWING TEST PROCEDURE Z-AXIS)



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MASS MOMENT OF INERTIA OF WOOD BOARD (PENDULUM SWING TEST RESULTS Z-AXIS)

Trial #	Duration for 5 Oscillations (s)	Average Time (s)
1	8.46	8.38
2	8.33	
3	8.35	

Radius to Center	Average Time Period Tavg(s)	MOI (kg*m^2)
0.535	1.67	0.119

MOI (Pendulum) (kg*m^2)	MOI (Torsional Vibration) (kg*m^2)	% Error
0.129	0.0998	19.2

Theoretical MOI	MOI (Pendulum)	MOI (Torsional Vibration)
0.0966	0.129	0.0998

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MASS MOMENT OF INERTIA OF A B-2 SPIRIT MODEL (TORSION TEST SETUP)

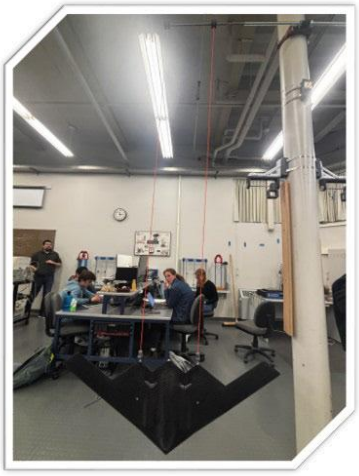


FIGURE: B2 HUNG TO TEST ROLL



FIGURE: B2 HUNG TO TEST YAW

MASS MOMENT OF INERTIA OF B-2 SPIRIT MODEL (TORSION TEST PROCEDURE ROLL AXIS)



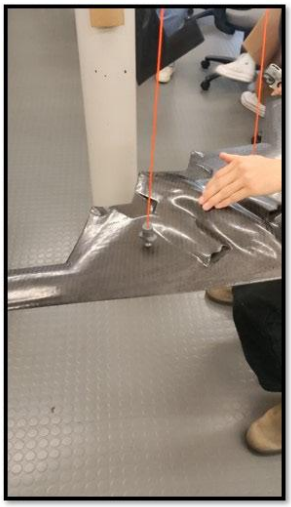
MASS MOMENT OF INERTIA OF B-2 SPIRIT MODEL (TORSION TEST RESULTS ROLL AXIS)

Axis of Rotation	Trial #	Duration for 5 Oscillations (s)	Average Time for 5 Oscillations (s)
Roll	1	19.51	19.62
	2	19.71	
	3	19.65	

Axis of Rotation	Cable Distance (m)	Height of Cables (m)	Average Time Period Tavg(s)	MOI (kg*m^2)
Roll	0.286	2	3.92	0.0726

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MASS MOMENT OF INERTIA OF B-2 SPIRIT MODEL (TORSION TEST PROCEDURE YAW AXIS)



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MASS MOMENT OF INERTIA OF B-2 SPIRIT MODEL (TORSION TEST RESULTS YAW AXIS)

Axis of Rotation	Trial #	Duration for 5 Oscillations (s)	Average Time for 5 Oscillations (s)
Yaw	1	21.48	21.61
	2	21.76	
	3	21.59	

Axis of Rotation	Cable Distance (m)	Height of Cables (m)	Average Time Period Tavg(s)	MOI (kg*m^2)
Yaw	0.286	2	4.322	0.0993

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MASS MOMENT OF INERTIA OF B-2 SPIRIT MODEL (PENDULUM SWING YAW TEST SETUP)

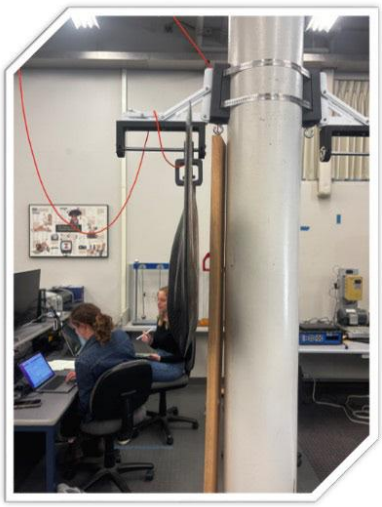


FIGURE: B2 HUNG TO TEST YAW (SIDE)

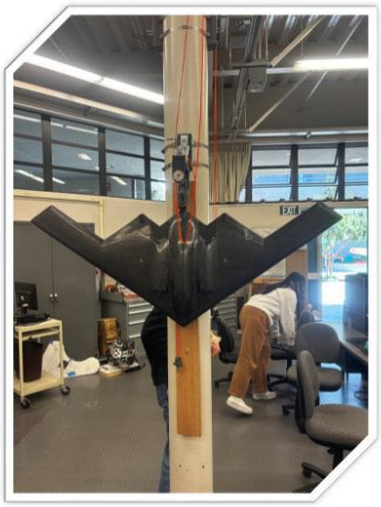
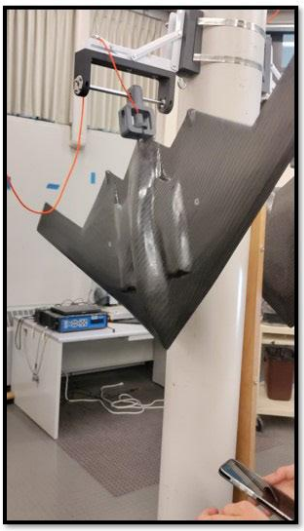


FIGURE: B2 HUNG TO TEST YAW (FRONT)

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MASS MOMENT OF INERTIA OF B-2 SPIRIT MODEL (PENDULUM SWING TEST PROCEDURE YAW AXIS)



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MASS MOMENT OF INERTIA OF B-2 SPIRIT MODEL (PENDULUM SWING RESULTS YAW AXIS)

Trial #	Duration for 5 Oscillations (s)	Average Time (s)
1	6.88	6.86
2	6.77	
3	6.90	

Radius to Center	Average Time Period Tavg(s)	MOI (kg*m^2)
0.2604	1.37	0.0998

MOI (Pendulum) (kg*m^2)	MOI (Torsional Vibration) (kg*m^2)	% Error
0.0998	0.0883	13

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CONCLUSION AND DISCUSSION

- TORSIONAL VIBRATION RESULTS CLOSELY MATCHED WOOD BEAM'S THEORETICAL VALUES WITH LOW ERROR (1.98% FOR Y, 3.27% FOR Z), CONFIRMING THE METHOD AND PROCEDURE ARE RELIABLE. HOWEVER, THE PENDULUM SWING METHOD PRODUCED LARGER ERRORS (25.8% FOR Y, 19.2% FOR Z).
- THE PENDULUM SWING DATA IS LIKELY AFFECTED BY DAMPING AND ALLIGNMENT ISSUES MUCH GREATER THAN THE TORSIONAL VIBRATION. THE FRICTION AT THE ROTATING AXLE MAY CONTRIBUTE THE MOST DAMPING.
- THE BOMBER HAS A COMPLEX GEOMETRIC SHAPE, THEREFORE OUR MEANS OF CALCULATING A THEORETICAL MOMENT OF INERTIA IS NOT FEASIBLE; FOR SUCH CASES WE CAN ONLY RELY ON EXPERIMENTAL DATA.
- THE PENDULUM SWING DATA IS ALSO RELATIVELY HIGHER THAN THE TORSIONAL VIBRATION DATA WITH A 13% ERROR BETWEEN THE TWO. OVERALL, THE RESULTS FROM THE PENDULUM SWING ARE CONSISTENTLY HIGHER.
- TORSIONAL VIBRATION IS THE MORE CONSISTENT METHOD TO MEASURE MOMENTS OF INERTIA. CONSIDERING ITS ACCURACY WITH THE THEORETICAL VALUES OF THE WOOD BEAM, IT IS PLAUSIBLE TO INFER THAT THE MOMENT OF INERTIA CALCULATED BY TORSIONAL VIBRATION IS CLOSE TO THE BOMBER'S TRUE MOI VALUE.
- USING A STRING ON A BALL BEARING INSTEAD OF A SOLID AXLE FOR THE PENDULUM SWING METHOD, AS WELL AS HAVING A MORE CONTROLLED RELEASE ON THE OBJECT CAN IMPROVE DATA RELIABILITY. A HIGH-SPEED CAMERA CAN HELP WITH TIMING OF THE PENDULUM. ADDITIONAL TRIALS COULDN'T HURT.



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